FINM3013 Introduction to Financial Derivatives

Chapter 9 Exercises

Solutions

1.

(a) No effect

(b) No effect

(c) The put option contract gives the right to sell 250 shares for $24 each

(d) The put option contract gives the right to sell 105 shares for 60/1.05 = $57.14

2.

(a) The margin requirement is the greater of 500×(10 + 0.2×58) = 10,800 and 500×(10+0.1×64) = 8,200. It is $10,800.

(b) The margin requirement is the greater of 500×(10+0.15×58) = 9,350 and 500×(10+0.1×64) = 8,200. It is $9,350.

(c) The margin requirement is the greater of 500×(10+0.2×70-6) = 9,000 and 500×(10+0.1×64) = 8,200. It is $9,000.

(d) No margin is required if the trader is buying

3. Figure 1 shows the way in which the investor's profit varies with the stock price in the first case. For stock prices less than $30 there is a loss of $1,200. As the stock price increases from $30 to $50 the profit increases from –$1,200 to $800. Above $50 the profit is $800. Figure 2 shows the way in which the profit varies with stock price in the second case. In this case the profit pattern has a zigzag shape. The problem illustrates how many different patterns can be obtained by including calls, puts, and the underlying asset in a portfolio.

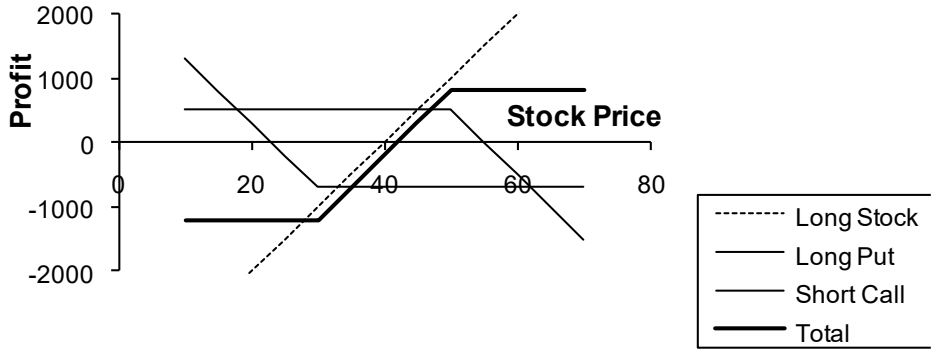


Figure 1 Profit for the first case

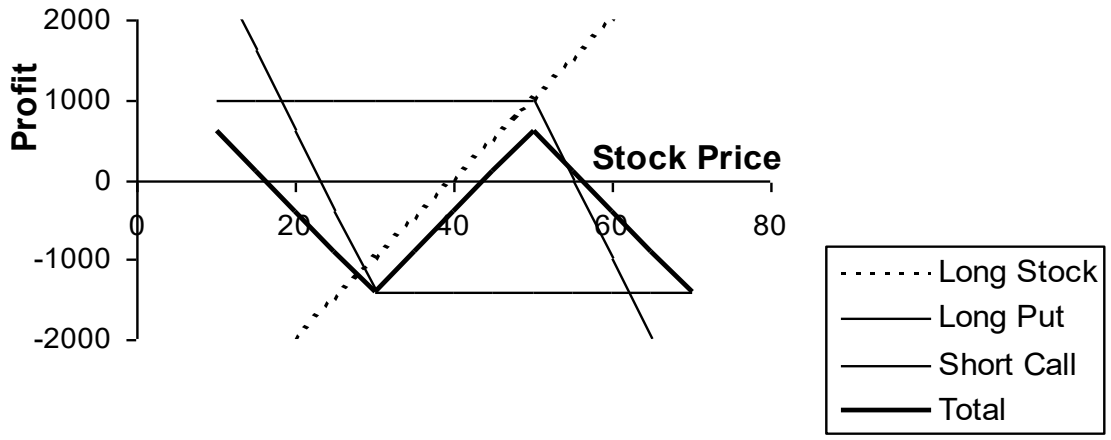


Figure 2 Profit for the second case